

KALAIYARASAN P

B.Tech Biotechnology Student — Aspiring Researcher

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Professional Summary

Dedicated B.Tech Biotechnology student (2023–2027) with practical laboratory experience in molecular biology, microalgae cultivation, environmental biotechnology, and clinical laboratory procedures. Skilled in DNA/RNA extraction, PCR/qPCR, microscopy, water quality analysis, biomass evaluation, and scientific documentation under Good Laboratory Practices (GLP). Passionate about clinical research, biotechnology innovation, and sustainable biological solutions.

Education

B.Tech Biotechnology	2023–2027
Muthayammal Engineering College, Rasipuram	
Status: Pursuing	
Higher Secondary Certificate (HSC)	2022–2023
Government Higher Secondary School	62.5%
Secondary School Leaving Certificate (SSLC)	2020–2021
Government Higher Secondary School	100%

Internship Experience

Research Intern	2025
Phycosol Ecotech Pvt. Ltd., Tamil Nadu	

- Cultivated and maintained microalgae under controlled laboratory conditions.
- Prepared nutrient media and monitored culture growth performance.
- Measured pH, EC, TDS, Optical Density (OD), biomass, and chlorophyll content.
- Contributed to projects involving CO₂ absorption, wastewater treatment, and biofertilizer applications.
- Maintained scientific records and documentation following GLP standards.

Laboratory Intern	2025
Sun Laboratory, Tamil Nadu	

- Assisted in clinical biochemistry procedures and biological sample processing.
- Participated in laboratory testing, documentation, and quality control activities.
- Recorded and verified laboratory data accurately.
- Followed laboratory safety regulations and Good Laboratory Practices (GLP).

Research Project

Nature-Derived Medium for Indoor Algal CO₂ Sequestration	2025
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- Developed a low-cost indoor microalgae cultivation system for carbon sequestration.

- Evaluated natural-source culture media and compared growth performance with commercial formulations.
- Monitored pH, EC, TDS, and Optical Density to assess growth efficiency.
- Conducted chlorophyll and biomass analysis to evaluate carbon capture potential.
- Documented research findings and laboratory observations using standard scientific practices.

Technical Skills

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|-------------------------------------|---------------------------------------|
| • PCR and qPCR Techniques | • Chlorophyll and Biomass Analysis |
| • DNA and RNA Extraction | • Spectrophotometry |
| • Agarose Gel Electrophoresis | • Media Preparation |
| • Microscopy and Sample Observation | • Experimental Documentation |
| • Microalgae Cultivation | • Good Laboratory Practices (GLP) |
| • Water Quality Analysis | • MS Office (Word, Excel, PowerPoint) |

Certifications & Training

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| • Good Clinical Practice (NIDA) | • RNA Isolation & qPCR Training |
| • Clinical Trial Management | • Cancer Cell Culture Techniques |
| • Experimental Biotechnology (NPTEL) | • Dot ELISA Workshop |
| • Stress Management (NPTEL) | • Smart India Hackathon 2026 |
| • QA/QC & Pharmaceutical Production | |

Research Interests

Clinical Research — Clinical Trials — Clinical Data Management — Algal Biotechnology — Environmental Biotechnology — Sustainable Bio-products — Food Technology

Achievements

- Participant – Smart India Hackathon 2026.
- Completed Internship Training in Algal Biotechnology at Phycosol Ecotech Pvt. Ltd.
- Completed multiple biotechnology and clinical research certification programs.